

Lecture 19- OSPF States,DR BDR,LSA Types,Network Types

1 Multicast Addresses of RIP, OSPF, and EIGRP

RIP Multicast Address

 **224.0.0.9**

Used by **RIPv2** to send routing updates to all RIP routers.

Protocol: Routing Information Protocol

OSPF Multicast Addresses

 **224.0.0.5** – All OSPF Routers

 **224.0.0.6** – All DR/BDR Routers

Protocol: Open Shortest Path First

EIGRP Multicast Address

 **224.0.0.10**

Protocol: Enhanced Interior Gateway Routing Protocol

Why Multicast is Used?

-  Reduces network load
 -  Sends updates only to routers
 -  More efficient than broadcast
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Real-Life Example

Type Example

Broadcast Shouting in a market 

Multicast Sending message in a WhatsApp group 

Unicast Calling one friend 

Multicast = smart communication 

2 What is Router ID (RID) in OSPF?

Router ID (RID) is a **32-bit number (looks like IP address)** that uniquely identifies a router in OSPF.

It is like a **name tag for routers** 

Example:

```
router ospf 1
router-id 1.1.1.1
```

After setting manually:

```
clear ip ospf process
```

Why Router ID is Important?

Purpose	Why Important
SPF Calculation	Root of shortest path tree 
LSA Origination	Identifies who created LSA
DR/BDR Election	Tie-breaker
Neighbor Relationships	Unique identity

Router ID Selection Order (Automatic)

- 1 Manually configured (highest priority)
- 2 Highest IP on loopback interface
- 3 Highest IP on active physical interface

💡 Loopback preferred because it never goes down.

Daily Life Example

Router ID = Signature on every official document 📝
Without signature → no identity

3 OSPF Neighborhood States (7 States)

OSPF routers go through 7 stages before full adjacency.

1 Down

No hello received ❌

2 Init

Hello received
But your Router ID not seen in neighbor hello

Unidirectional communication ⚠️

3 2-Way

Both routers see each other's Router ID ✅

DR/BDR election happens here 👑

Full adjacency formed:

- Point-to-point

- With DR/BDR in broadcast networks

ExStart

Master/Slave election 

Higher Router ID = Master

5 **Exchange**

Routers exchange Database Description (DBD) packets 

They compare LSDB summaries.

6 **Loading**

Missing LSAs requested (LSR)

Neighbor replies with LSU

Database synchronization happening 

7 **Full**

Final stage 

LSDB fully synchronized.

State Flow

Down → Init → 2-Way → ExStart → Exchange → Loading → Full

Verify Command

show ip ospf neighbor

DR, BDR, DR-Other

Used in multi-access networks like Ethernet.

Without DR/BDR:

10 routers = 45 adjacencies 😞

With DR/BDR:

Only DR & BDR form full adjacency

Much less overhead 👍

Definitions

Role	Meaning
DR	Main router sharing LSAs
BDR	Backup router
DR Other	All other routers

DR/BDR Election Process

-  Highest OSPF priority wins
(Default priority = 1)
 -  If tie → Highest Router ID wins
 -  Priority 0 → Not eligible
- ⚠ No preemption

Example

```
interface f0/0  
ip ospf priority 100
```

Hello & Dead Timer

Hello = every 10 sec

Dead = 40 sec

If no hello for 40 sec → neighbor down ❌

Simple Understanding

Highest loopback IP → DR

Second highest → BDR

Others → DROTHER

DR job = Share LSAs to everyone 🗨️

5 What is LSA?

LSA = Link State Advertisement

It is a message that routers send to share network information.

All routers collect LSAs → build LSDB → run SPF.

Types of LSAs

◆ Type 1 – Router LSA

Generated by every router

Inside same area only

“I am Router1 and these are my links.”

◆ Type 2 – Network LSA

Generated by DR

Lists all routers in LAN

◆ Type 3 – Summary LSA

Generated by ABR

Shares routes between areas

◆ Type 4 – ASBR Summary

Tells how to reach ASBR

◆ Type 5 – External LSA

Generated by ASBR

Advertises external routes

◆ Type 7 – NSSA External

Used in NSSA areas

Converted to Type 5 by ABR

Easy Analogy

City = OSPF Area

Type 1 → Each house shares details

Type 2 → Society leader shares member list

Type 3 → City shares info to other city

Type 5 → Border gate shares outside world info



OSPF Algorithm (SPF)

OSPF uses **Dijkstra Shortest Path First Algorithm**

Algorithm: Dijkstra's algorithm

Steps

- 1 Build LSDB
- 2 Run SPF
- 3 Install best routes in routing table

Metric = Cost (based on bandwidth)

Lower cost = better path 🚀



Real-Life Example

Google Maps choosing shortest route 🗺️

Home = Root

Different roads = Different costs

SPF picks best road.



OSPF Areas

Small network → Single area OK

Large network → Must divide into areas

Why?

- ✗ Large LSDB
 - ✗ More memory
 - ✗ More CPU
 - ✗ More SPF recalculations
-

Backbone Area

Area 0 = Backbone

All areas must connect to Area 0

Router Types

Router Type	Meaning
Internal Router	All interfaces same area
ABR	Connected to multiple areas
Backbone Router	Connected to Area 0

Route Types

Intra-area → Same area
Inter-area → Different area

Rules

- ✓ Areas must be continuous
 - ✓ All areas connect to Area 0
 - ✓ Same subnet → same area
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OSPF Neighbor Requirements

Must Match:

- ✓ Area number
- ✓ Same subnet
- ✓ OSPF process running
- ✓ Unique Router ID
- ✓ Hello & Dead timers
- ✓ Authentication

✓ MTU

✓ Network type

If MTU mismatch → Neighbor forms but routing fails ⚠



OSPF Network Types

1 Broadcast

Example: Ethernet

✓ Multicast supported

✓ DR/BDR election

2 NBMA

Example: Frame Relay

✗ No multicast

✓ Manual neighbor config

✓ DR/BDR used

3 Point-to-Point

Example: Serial links

✓ No DR/BDR

✓ Only 2 routers

Simple and efficient 👍



What is HDLC?

HDLC = High-Level Data Link Control

Layer 2 protocol

Used on serial links

Ensures:

- ✓ Frame identification
- ✓ Error detection (CRC)
- ✓ Proper delivery

Cisco HDLC is proprietary.



Daily Life Example

IP packet = Letter 

Envelope = Encapsulation

Stamp = Error check

Courier = Serial link

Encapsulation = Wrapping data before sending 



Final Quick Revision

- ✓ Multicast addresses
 - ✓ Router ID selection
 - ✓ 7 OSPF states
 - ✓ DR/BDR election
 - ✓ LSA types
 - ✓ SPF algorithm
 - ✓ Areas & ABR
 - ✓ Neighbor requirements
 - ✓ Network types
 - ✓ HDLC
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CHAPTER SUMMARY

1 Multicast Addresses (RIP, OSPF, EIGRP)

Protocol Multicast Address

RIP v2 224.0.0.9

OSPF 224.0.0.5 (All OSPF Routers)

OSPF 224.0.0.6 (All DR/BDR)

EIGRP 224.0.0.10

Why Multicast?

- Reduces network traffic 
- Sends updates only to routers (not PCs)
- Efficient communication

2 Router ID (RID)

- 32-bit unique number (looks like IP)
- Identifies router in OSPF
- Used in SPF, LSA creation & DR election 

Selection Order:

1. Manual configuration
2. Highest loopback IP
3. Highest physical interface IP

Loopback preferred because it never goes down 

3 OSPF Neighbor States (7 States)

Down → Init → 2-Way → ExStart → Exchange → Loading → Full

State	Meaning
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Down	No Hello received
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Init	Hello received (one-way)
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2-Way	Bidirectional communication
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ExStart	Master/Slave decided
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Exchange	DBD packets exchanged
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Loading	LSR/LSU packets exchanged
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Full	LSDB synchronized 🎉
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DR / BDR

Used in Broadcast & NBMA networks.

Why?

Without DR → Too many adjacencies 😞

With DR → Reduced overhead 👍

Election:

1. Highest priority
 2. Highest Router ID
 3. Priority 0 = Not eligible
- ⚠️ No preemption
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LSA Types

Type Name	Generated By
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1	Router LSA	Every router
2	Network LSA	DR
3	Summary LSA	ABR
4	ASBR Summary	ABR
5	External	ASBR
7	NSSA External	ASBR (NSSA)

LSAs → Build LSDB → Run SPF 🧠



SPF Algorithm

- Uses Dijkstra Algorithm
- Metric = Cost (based on bandwidth)
- Lower cost = Better path 🚀



OSPF Areas

- Area 0 = Backbone
- All areas must connect to Area 0
- Reduces LSDB size & CPU usage

Router Types:

- Internal Router
 - ABR
 - Backbone Router
 - ASBR
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8 Neighbor Requirements

Must match:

- ✓ Area
 - ✓ Subnet
 - ✓ Timers
 - ✓ Authentication
 - ✓ MTU
 - ✓ Network Type
 - ✓ Unique RID
-

Network Types

Type	DR/BDR?	Example
Broadcast	Yes	Ethernet
NBMA	Yes	Frame Relay
Point-to-Point	No	Serial

HDLC

- Layer 2 protocol
 - Used on serial links
 - Error detection via CRC
 - Cisco HDLC = proprietary
-

CONCLUSION

OSPF is:

- ✓ Link-state protocol
- ✓ Uses LSAs
- ✓ Builds LSDB
- ✓ Runs SPF algorithm
- ✓ Uses Areas for scalability
- ✓ Elects DR/BDR in multi-access networks

👉 OSPF is scalable, efficient, and enterprise-level routing protocol.

👉 Most commonly used IGP in real networks 🌐

Q & A

1 What is OSPF?

OSPF (Open Shortest Path First) is a link-state routing protocol that uses Dijkstra SPF algorithm to calculate shortest path based on cost metric.

It sends LSAs instead of routing table updates like RIP.

2 Why OSPF is better than RIP?

RIP

Distance vector

Hop count metric

Max 15 hops

Slow convergence

OSPF

Link-state

Cost metric

No hop limit

Fast convergence 🚀

3 What is Router ID?

Router ID is a 32-bit number used to uniquely identify a router in OSPF.

Used in:

- LSA creation
 - DR election
 - SPF calculation
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4 What is DR and why needed?

DR (Designated Router) reduces number of adjacencies in multi-access network.

Without DR → $n(n-1)/2$ adjacencies

With DR → Only DR & BDR full adjacency

5 What is BDR?

Backup Designated Router.

Takes over if DR fails ⚡

6 What happens in 2-Way state?

Bidirectional communication confirmed.

DR/BDR election happens here 👑

7 What is LSA?

Link State Advertisement — contains information about links & topology.

8 What is LSDB?

Link State Database — collection of all LSAs.

9 What is SPF?

Shortest Path First algorithm used to calculate best path.

10 What is OSPF Cost?

$\text{Cost} = \text{Reference bandwidth} / \text{Interface bandwidth}$

Default reference = 100 Mbps

1 1 Explain all OSPF states.

(Explain all 7 states clearly like above section.)

1 2 What are different LSA types?

Explain Type 1 to 7 in detail.

1 3 What is ABR?

Area Border Router — connects multiple areas.

Generates Type 3 LSAs.

1 4 What is ASBR?

Autonomous System Boundary Router — injects external routes.

Generates Type 5 LSAs.

1 5 What is Backbone Area?

Area 0.

All areas must connect to it.

1 6 What is MTU mismatch issue?

Neighbors may form but stuck in Exchange state.

Routing may fail ⚠️

1 7 Why loopback preferred for Router ID?

Because loopback never goes down.

1 8 What is Hello & Dead timer?

Hello = 10 sec

Dead = 40 sec

If no hello in 40 sec → neighbor down.

1 9 What is difference between Broadcast & NBMA?

Broadcast supports multicast.

NBMA doesn't support multicast — needs manual neighbor config.

2 0 What is OSPF network types?

Broadcast, NBMA, Point-to-Point.

2 1 Why OSPF is called link-state protocol?

Because routers exchange link-state information instead of full routing tables.

2 2 What is difference between 2-Way and Full state?

2-Way → Only neighbor visibility

Full → LSDB fully synchronized

2 3 Why DR election does not preempt?

To avoid instability and frequent topology changes.

2 4 What is NSSA?

Not-So-Stubby Area.

Allows external routes as Type 7 LSAs.

2 5 What command verifies OSPF neighbors?

show ip ospf neighbor

2 6 How to manually set Router ID?

router ospf 1
router-id 1.1.1.1
clear ip ospf process

2 7 What is difference between Type 3 and Type 5 LSA?

Type 3 → Inter-area

Type 5 → External routes

2 8 What happens if Area mismatch?

Neighbor will not form ❌

2 9 What is OSPF metric calculation formula?

Cost = 100,000,000 / Bandwidth (in bps)

3 0 What is difference between Internal and Backbone router?

Internal → All interfaces same area

Backbone → Connected to Area 0

3 1 Neighbor stuck in ExStart — reason?

- MTU mismatch
 - Router ID conflict
 - Network type mismatch
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3 2 DR fails — what happens?

BDR becomes DR immediately ⚡

3 3 Two routers same priority — who becomes DR?

Highest Router ID wins 🏰

3 4 Why OSPF converges faster?

Because it sends triggered updates and calculates full topology.

3 5 How to prevent router from becoming DR?

ip ospf priority 0
